Fatima Jinnah Women University

Opening Portals of Excellence Through Higher Education

Assignment 01

Git, Gitea, GitHub, Git LFS, and GitHub Pages

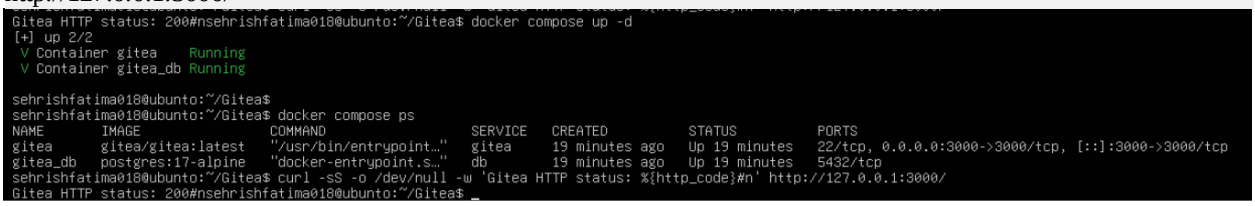
Student name	Sehrish Fatima
Registration number	FA24B1-SE-065
Section	A
Course	Cloud Computing
GitHub username	Sehrishfatima018
Ubuntu prompt	Sehrishfatima018@ubuntu:~$
Deadline	10 October 2026

Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Step 1.1: Install and Run Gitea on the Ubuntu Server

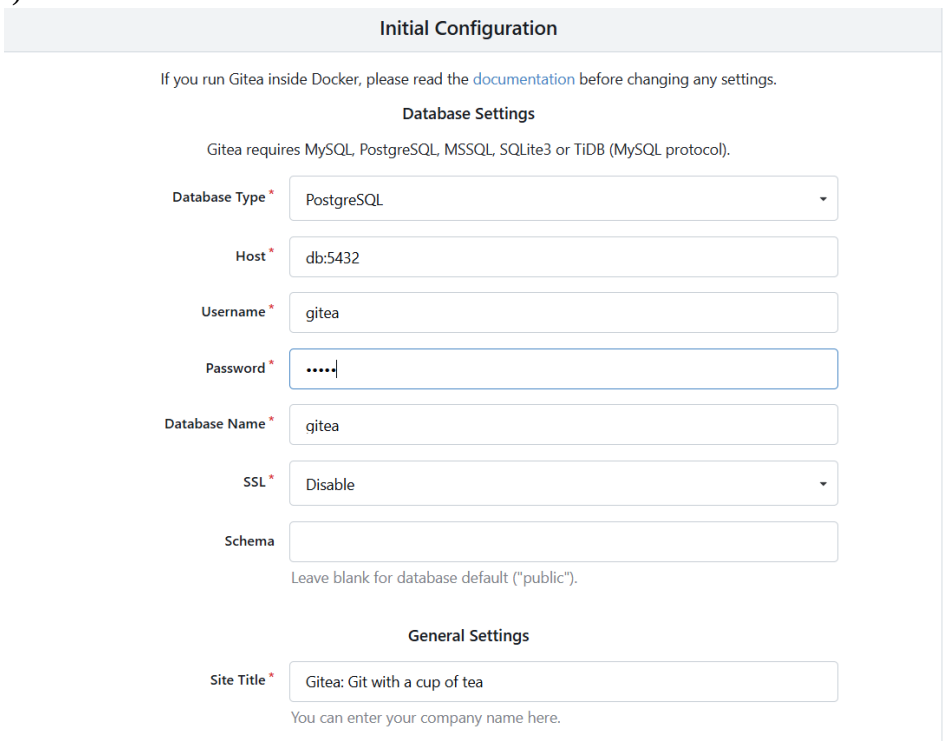
Installed Docker and the Compose plug-in, cloned the instructor's Gitea repository, started Gitea and PostgreSQL, and confirmed Gitea answers on port 3000.

```
git clone https://github.com/WaqasSaleem97/Gitea.git
docker compose up -d docker
compose ps
curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n"
http://127.0.0.1:3000/
```



task1_gitea_running.png: both containers running and Gitea HTTP status 200

Gitea installation page, using the required PostgreSQL database settings (host db:5432, user gitea, database gitea).



Gitea installation page: database settings

Administrator account created to complete the installation.

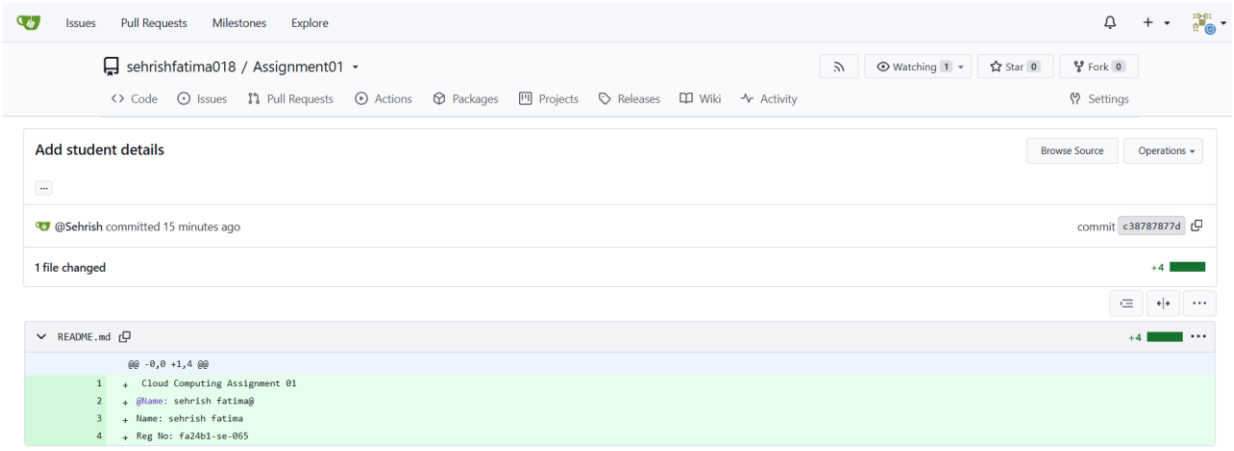
Step 1.2: Create and Push the Initial Repository

Created an empty public Gitea repository named Assignment01. On the server, created a separate local repository with a README containing my name and registration number, added the gitea remote without a token in the URL, and pushed with a personal access token entered at the password prompt.



task1_gitea_push.png: gitea remote and successful push to main

Step 1.3: Verify the Gitea Repository



task1_gitea_repository.png: public repository with rendered README

Task 2: Push the Same Repository to GitHub

Step 2.1: Create the GitHub Repository and Verify Both Remotes

Created an empty public GitHub repository named Assignment01 and added it as the github remote.

```
git remote add github https://github.com/ayeshaaafifa/Assignment01.git git remote -v
```

```
sehrishfatima018@ubunto:~/Assignment01$ cd ~/Assignment01
sehrishfatima018@ubunto:~/Assignment01$ git remote add github https://github.com/sehrishfatima018/Assignment01.git
sehrishfatima018@ubunto:~/Assignment01$ git remote -v
gitea   http://sehrishfatima018@127.0.0.1:3000/sehrishfatima018/Assignment01.git (fetch)
gitea   http://sehrishfatima018@127.0.0.1:3000/sehrishfatima018/Assignment01.git (push)
github  https://github.com/sehrishfatima018/Assignment01.git (fetch)
github  https://github.com/sehrishfatima018/Assignment01.git (push)
```

task2_remotes.png: gitea and github remotes, no token or password in the URLs

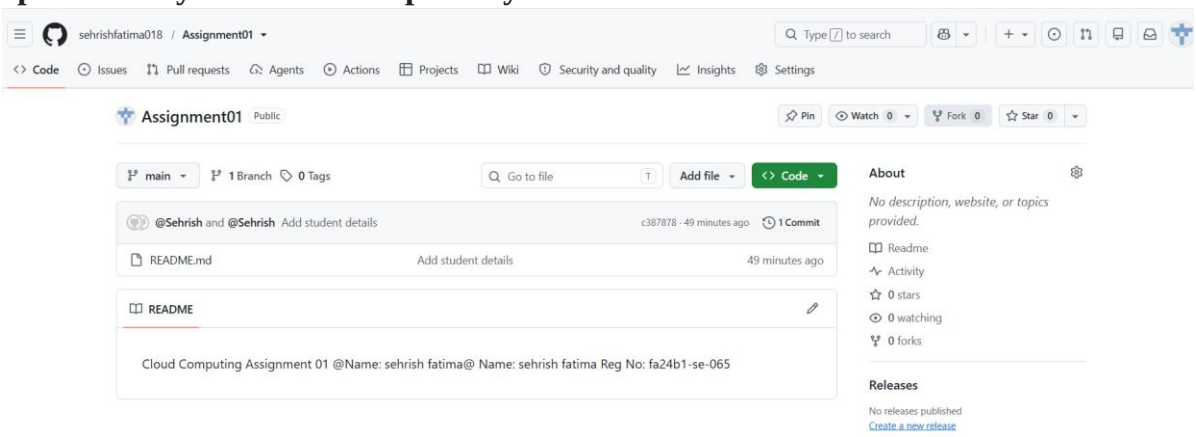
Step 2.2: Push to GitHub

```
git push -u github main
```

```
sehrishfatima018@ubunto:~/Assignment01$ git push -u github main
Username for 'https://github.com': sehrishfatima018
Password for 'https://sehrishfatima018@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 524 bytes | 52.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sehrishfatima018/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
sehrishfatima018@ubunto:~/Assignment01$
```

task2_github_push.png: successful push of main to GitHub

Step 2.3: Verify the GitHub Repository

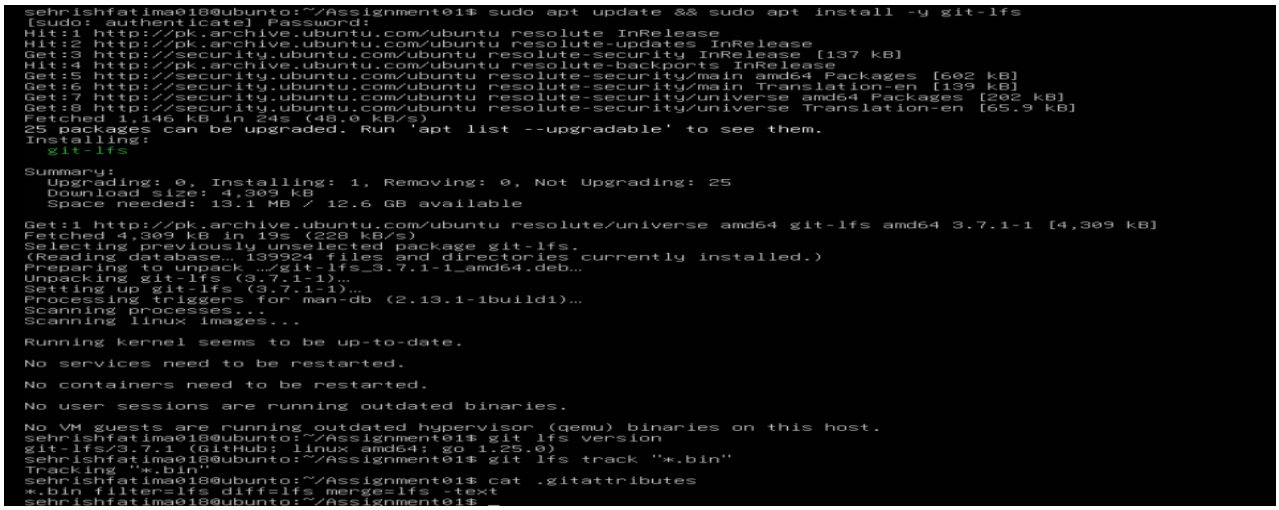


task2_github_repository.png: public repository with rendered README

Task 3: Track Three Large Files with Git LFS

Step 3.1: Configure Git LFS

```
git lfs install git lfs version git
lfs track "*.bin" cat
.gitattributes
```

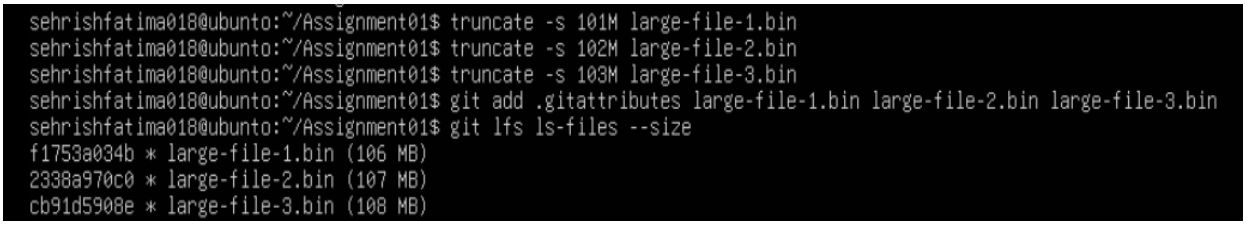


*task3_lfs_setup.png: Git LFS initialized, version shown, *.bin rule in .gitattributes*

Step 3.2: Create and Verify Three LFS Files

Created three files of 101 MiB, 102 MiB and 103 MiB, staged them, and listed them with Git LFS (shown as 106, 107 and 108 MB).

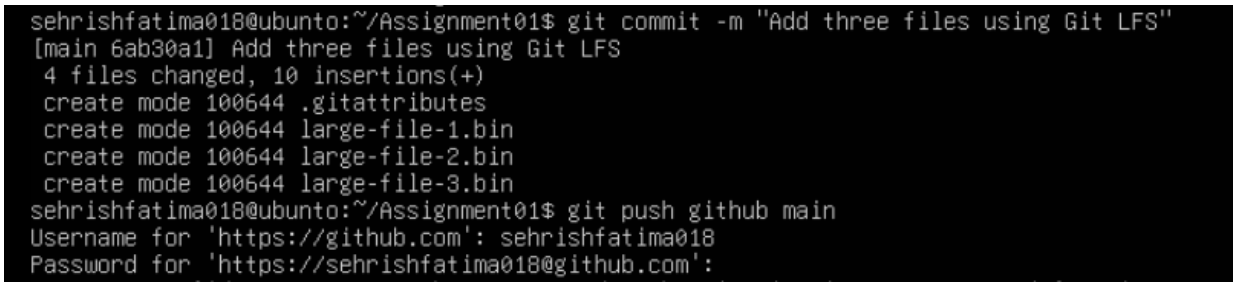
```
truncate -s 101M large-file-1.bin truncate -s 102M large-file-
2.bin truncate -s 103M large-file-3.bin
git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
git lfs ls-files --size
```



task3_lfs_files.png: three files tracked by Git LFS, each larger than 100 MiB

Step 3.3: Commit and Push the LFS Files

```
git commit -m "Add three files using Git LFS" git push github main
```



task3_lfs_push.png: LFS upload 100% (3/3) and successful push to GitHub